WEEK 10 PROJECT REPORT

King County House Sales Interactive Dashboard & Advanced Analytics

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1. Introduction

This project shows how we built the final interactive analytics dashboard for the King County House Sales dataset using Microsoft Power BI Desktop. Following the data cleaning and column creation completed in Week 9, the main goal for Week 10 is to create business calculations (DAX measures) and display them using interactive visuals.

The dashboard provides real-estate managers with a complete look at house sales trends. It features key performance cards, a geographic location map, charts showing what impacts house prices (like water views, bedrooms, and structural grades), and a dynamic slider filter. The entire dashboard connects together, allowing users to slice and filter data instantly.

2. Objectives

• Formulate 5 business- DAX measures for core real-estate metrics.

• Organize all metrics into a clean, dedicated calculation table called Measures.

• Design a top KPI header row using Card visuals to show summary numbers clearly.

• Build a geographic Map visual using Latitude and Longitude to display property locations across Seattle.

• Create a Clustered Bar Chart to analyze the massive price premium of waterfront homes.

• Create a Clustered Column Chart to display how the number of bedrooms affects pricing.

• Create a Line Chart to chart how house pricing goes up alongside structural quality grades.

• Establish a dynamic Numeric Range Parameter slider to let users filter home square footage in real time.

3. Tools and Technologies Used

Tool	What we used it for
Microsoft Power BI Desktop	The main application used to write DAX math formulas and arrange the visual dashboard interface.
Data Analysis Expressions (DAX)	The formulas written to calculate counts, averages, and medians on the live dataset.
Numeric Range Parameters	The feature used to build an interactive query slider scale widget on the page.

4.Formulating Advanced DAX Measures

We created a separate table named Measures to store all calculations neatly. We then clicked New Measure to create the 5 distinct calculations .

1.Total Properties Sold

```
Total Properties Sold = COUNT('kc_house_data'[id])
```

2. Average House Price

```
Avg House Price = AVERAGE('kc_house_data'[price])
```

3. Median Price

```
Median Price = MEDIAN('kc_house_data'[price])
```

4. Average Living Area

```
Avg Living Area = AVERAGE('kc_house_data'[sqft_living])
```

5. Average Price per Square Foot

```
Avg Price Per SqFt = AVERAGE('kc_house_data'[Price per sqft])
```

5.Building the Geographic Map Visual

We dragged a blank Map visual onto the middle of the screen. We navigated to the column settings options and configured the fields:

• Latitude: Loaded with the lat database column.

• Longitude: Loaded with the long database column.

• Color: Bound to the Avg House Price calculation measure.

6.Dashboard pages

We created three specialized charts beneath the map layout area to show the primary drivers of real-estate pricing:

• Waterfront Status Premium (Bar Chart): Shows the massive value difference between waterfront vs landlocked properties using waterfront on the Y-axis and Avg House Price on the X-axis.

• Bedroom Counts Trend (Column Chart): Details home costs broken down by size using bedrooms on the X-axis and Avg House Price on the Y-axis.

• Quality Grade Impacts (Line Chart): Charts the steady upwards line showing how average price scales with better construction quality grades using grade on the X-axis and Avg House Price on the Y-axis.

• Market Remodeling Share (Donut Chart): Tracks the percentage distribution of flipped or updated inventory by using the custom column Is Renovated in the Legend and Total Properties Sold in the Values.

• Property Age vs. Valuation Trajectory (Line Chart): Charts how the structural lifespan of a property impacts its final transaction cost using Property Age at Sale on the X-axis and Avg House Price on the Y-axis.

• Affordability Volume Segments (Clustered Column Chart): Displays absolute supply volumes available to different homebuyer profiles using the custom column Price Tier (Budget, Mid-Range, High-End) on the X-axis and Total Properties Sold on the Y-axis.

7.Implementing Dynamic Parameter Slider Widget

We navigated to the Modeling tab menu options and selected New Parameter > Numeric Range to satisfy the dynamic range filtering requirement.

• Name: Square Footage Filter

• Data Type: Whole number

• Minimum / Maximum: 200 to 10000

• Increment / Default: 100 step intervals starting at 200 base value.

Parameters

Add parameters to visuals and DAX expressions so people can use slicers to adjust the inputs and see different outcomes. Learn more

What will your variable adjust?

Numeric range

Name

Square Footage Filter

Data type

Whole number

Minimum

200

Maximum

10000

Increment

100

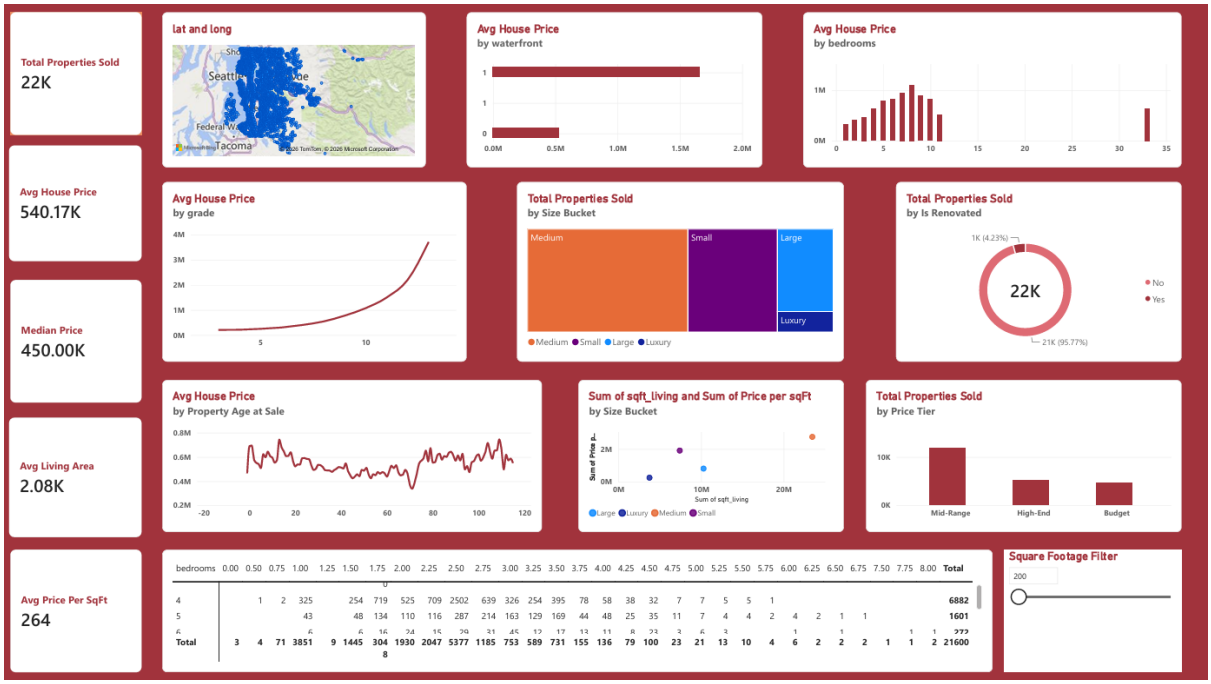
Default

200

Add slicer to this page

Create Cancel

Final Dashboard



8. Conclusion

In Week 10, we successfully completed the King County House Sales dashboard project in Power BI. We turned our raw dataset into an interactive report. By writing custom math formulas (DAX measures) and building several dashboard pages, we created a professional tool that makes analyzing the real estate market very easy.

Users can now open this project to instantly view home prices on a geographic map, see how water views and bedroom counts change a house's value, check how house age impacts cost, and use an interactive slider to filter properties by their square footage size. Every chart connects with the others, making the entire report fully interactive.